Post-Training an LLM for Complex Alignment

Raghu Mysore Shantharam[1]

[1]**Raghums534@gmail.com**

ABSTRACT

In this work, we investigate post-training for enforcing complex, compositional behavioral constraints in a small open-weight language model. We use a custom dataset constructed from the rules defined in the assignment and fine-tune Qwen2.5-0.5B-Instruct using supervised fine-tuning (SFT) with LoRA. We evaluate the model on 25 held-out cases using rule-level and case-level compliance metrics. Compared with the untuned baseline, SFT improves rule-level compliance from 33.33% to 58.59% and case-level compliance from 4.00% to 12.00%. Improvements occur for both individual and multi-rule cases, although no multi-rule case achieves complete compliance after SFT. The results demonstrate that LoRA-based SFT substantially improves protocol adherence while highlighting the difficulty of simultaneously satisfying multiple behavioral constraints. Additionally, we also notice much better performance with larger and models such as Qwen3-8B.

Keywords: LLM, Post-Training, SFT, LoRA, Rules, Qwen, Rule-level, Case-level

INTRODUCTION

Large language models (LLMs) can be adapted through post-training to follow task-specific behavioral requirements. This work investigates whether supervised fine-tuning (SFT) can improve adherence to a complex, compositional behavior protocol consisting of multiple trigger-dependent rules. We construct a custom dataset from the protocol and fine-tune Qwen2.5-0.5B-Instruct using parameter-efficient LoRA. The model is evaluated on 25 held-out cases using a rule-based evaluator that measures both rule-level and case-level compliance. We compare the untuned Qwen2.5-0.5B-Instruct model with its SFT counterpart and use Qwen3-8B as an additional baseline for model-scale comparison.

DATASET

We constructed a custom supervised fine-tuning dataset from the behavioral protocol specified in the assignment. The protocol contains 18 trigger-dependent rules, covering requirements such as response formatting, lexical constraints, conditional content, and interactions between multiple rules. Training examples were synthetically constructed as prompt–completion pairs, with examples designed to activate different rules and combinations of rules. The resulting training set contains 900 examples and was used exclusively for supervised fine-tuning.

A separate validation set was constructed to evaluate the model's ability to generalize the learned behaviors beyond the training examples. We selected 25 held-out use cases covering both individual-rule and multi-rule scenarios. The cases were designed by considering the trigger conditions of the protocol and selecting prompts that activate different rules as well as combinations of rules. This allows the evaluation to measure not only whether the model can follow individual constraints, but also whether it can satisfy several simultaneously triggered constraints.

The validation cases were kept separate from the training data and were evaluated using the same rule-based evaluator for both the untuned baseline and the SFT model. This provides a controlled comparison of the effect of post-training on protocol adherence.

EXPERIMENTAL SETUP

Primary LLM: Qwen2.5-0.5B-Instruct
Additional baseline: Qwen3-8B

GPU: NVIDIA Tesla T4
Platform: Google Colab
AI Assistants: ChatGPT and Claude

METHODS AND MATERIALS

Baseline Evaluation

We first evaluated the behavior of the untuned language models before applying any post-training. A set of 25 held-out validation cases was constructed from the behavioral protocol, covering both individual-rule and multi-rule scenarios. For each case, the user query was presented to the model together with the corresponding protocol rules and their trigger conditions. This allowed us to measure whether the baseline models could correctly apply the specified constraints without additional fine-tuning.

The baseline evaluation was performed using Qwen2.5-0.5B-Instruct and Qwen3-8B. The same validation cases and rule-based evaluation procedure were used to measure the responses. Each response was evaluated against every rule triggered by the corresponding input. We report both rule-level compliance, which measures the proportion of triggered rules satisfied, and case-level compliance, which requires all applicable rules in a case to be satisfied.

Supervised Fine-Tuning

We selected Qwen2.5-0.5B-Instruct as the primary model for the controlled post-training experiment. A custom supervised fine-tuning dataset containing 900 prompt–completion examples was constructed from the behavioral protocol. The examples were designed to expose the model to the different trigger conditions and response requirements specified by the 18 protocol rules, including examples involving multiple simultaneously active rules.

Parameter-efficient supervised fine-tuning was performed using Low-Rank Adaptation (LoRA), allowing trainable low-rank adapters to be introduced into selected attention projection layers while keeping the original model weights frozen. The LoRA configuration used a rank of 16, an α value of 32, and a dropout rate of 0.05, with the query, key, value, and output projection layers targeted for adaptation.

The model was trained for three epochs with a per-device batch size of 2 and gradient accumulation over four steps. A learning rate of 2×10^{-4} was used. Training was performed using mixed-precision FP16 computation on an NVIDIA Tesla T4 GPU. The final trained LoRA adapter was saved separately from the base model.

After training, the adapter was used with the original Qwen2.5-0.5B-Instruct model for inference. Importantly, the 25 validation cases were not used during training. The fine-tuned model was evaluated on exactly the same held-out cases and under the same evaluation procedure as the baseline model, allowing the effect of SFT to be measured under controlled conditions.

Evaluation Framework

To evaluate protocol adherence, we implemented a rule-based evaluator based directly on the 18 rules specified in the assignment. Each validation case was associated with the protocol rules expected to be triggered by its input. The evaluator first extracted the model's final response and then evaluated every triggered rule independently. For each rule, the evaluator records whether the response satisfies the corresponding behavioral requirement and provides a reason for the decision.

Two complementary metrics were used. Rule-level compliance is the proportion of triggered rule instances that are satisfied across the validation set. In contrast, case-level compliance is stricter: a case is considered successful only when all rules triggered by that case are satisfied. This distinction is particularly important for the protocol because multiple rules may be active simultaneously and must be satisfied together.

The 25 validation cases were divided into individual-rule and stacked multi-rule cases. This allows the evaluation to distinguish the ability to follow individual behavioral constraints from the ability to compose several constraints within a single response. The same evaluator, validation cases, and scoring procedure were applied to both the untuned and SFT models.

RESULTS

The models were evaluated on the held-out validation cases using the rule-based evaluation framework described above. For each case, the evaluator identified the protocol rules triggered by the input and

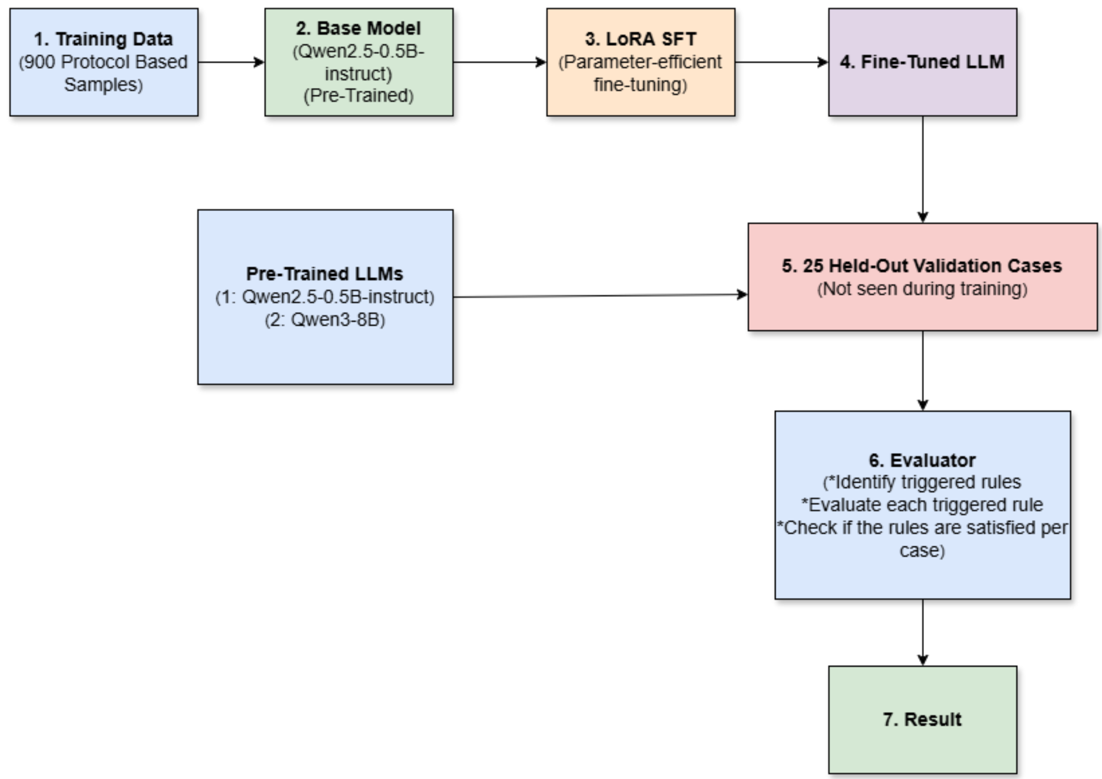

Figure 1. Block diagram of the experimental pipeline used for supervised fine-tuning and evaluation.

evaluated the generated response against each triggered rule independently. Across the 25-case evaluation set, 99 rule instances were triggered. This produced two complementary measures of performance: rule-level compliance and case-level compliance.

Rule-level compliance measures the proportion of individual triggered rules that were satisfied. Case-level compliance is stricter and considers a case successful only when all rules triggered by that case are satisfied. Thus, a model can achieve relatively high rule-level compliance while obtaining a lower case-level score if it frequently violates at least one requirement in multi-rule cases.

Overall Comparison

Table 1 summarizes the results obtained for the three model configurations.

Table 1. Protocol compliance across the evaluated models.

Model	Training	Rule-level	Case-level	Passed cases
Qwen2.5-0.5B	None	33.33%	4.00%	1/25
Qwen2.5-0.5B	LoRA SFT	**58.59%**	**12.00%**	**3/25**
Qwen3-8B	None	72.04%	56.52%	13/23

The controlled comparison between the Qwen2.5-0.5B baseline and its fine-tuned counterpart shows a substantial improvement following SFT. Rule-level compliance increased from 33.33% to 58.59%, corresponding to an increase of 25.26 percentage points. The number of completely compliant cases increased from 1 to 3 out of 25, raising case-level compliance from 4.00% to 12.00%. The SFT model therefore satisfied 58 of the 99 triggered rule instances, compared with 33 for the baseline.

The Qwen3-8B baseline achieved higher observed compliance than either Qwen2.5-0.5B configuration. This provides additional evidence that model scale can influence protocol-following ability. However, this comparison is presented as a reference rather than a strictly controlled experiment because the recorded Qwen3-8B evaluation contained 23 evaluated cases and 93 triggered rule instances.

Individual and Multi-Rule Cases

To investigate the compositional nature of the protocol, the validation cases were further divided into individual-rule and stacked/multi-rule cases. The individual-rule subset contains 17 cases, while the stacked subset contains 8 cases.

Table 2. Compliance by case type for Qwen2.5-0.5B before and after SFT.

Case type	Metric	Baseline	SFT
Individual	Rule-level	34.00%	**60.00%**
Individual	Case-level	5.88%	**17.65%**
Stacked	Rule-level	32.65%	**57.14%**
Stacked	Case-level	0.00%	0.00%

SFT improves rule-level compliance for both types of cases. Individual-rule compliance increases from 34.00% to 60.00%, while stacked-rule compliance increases from 32.65% to 57.14%. However, no stacked case is completely compliant after SFT. This indicates that the model learns a substantial portion of the individual behavioral constraints but continues to struggle when several constraints must be satisfied simultaneously. Figure 1, shows the basic block diagram of the experimental setup that was used during this assignment.

CONCLUSION

This work investigated whether supervised fine-tuning could improve adherence to a complex, compositional behavioral protocol in a small open-weight language model. Using a custom dataset of 900 examples, Qwen2.5-0.5B-Instruct was fine-tuned with LoRA and evaluated on 25 held-out cases using a rule-based compliance framework.

The controlled comparison showed a clear improvement following SFT. Rule-level compliance increased from 33.33% for the untuned model to 58.59% after SFT, while case-level compliance increased from 4.00% to 12.00%. Improvements were observed for both individual and stacked rule conditions. However, no stacked case achieved complete compliance after SFT, indicating that simultaneously satisfying multiple behavioral constraints remains challenging for the small model.

A practical limitation was encountered when investigating a QLoRA-based alternative. The available NVIDIA Tesla T4 environment introduced memory and mixed-precision compatibility constraints during the QLoRA training setup, preventing a stable end-to-end comparison with the LoRA configuration. We therefore used standard LoRA for the controlled SFT experiment. This limitation also highlights the practical importance of considering hardware constraints when selecting a post-training method.

Overall, the experiments demonstrate that parameter-efficient SFT can substantially improve protocol adherence in a small LLM, while the evaluation also exposes the remaining difficulty of compositional rule following.

FURTHER IMPROVEMENTS

Several directions could be explored to improve and extend the results obtained in this study. First, a larger and more diverse supervised fine-tuning dataset could be constructed, with greater coverage of individual rules, trigger conditions, and especially combinations of simultaneously active rules. This could help address the difficulty observed in stacked multi-rule cases.

Second, the evaluation benchmark could be expanded beyond the current 25 held-out cases. A larger and more systematically balanced set of validation cases could provide more reliable estimates of model performance and include a wider range of individual and multi-rule combinations.

Third, QLoRA could be investigated as an alternative parameter-efficient fine-tuning approach. Although a stable QLoRA experiment could not be completed under the available hardware and software configuration, a more suitable computational setup could enable a direct comparison between LoRA and QLoRA under otherwise identical conditions.

Finally, with access to greater computational resources, larger open-weight models could be fine-tuned and evaluated using the same dataset and evaluation framework. This would help investigate whether the remaining compositional limitations are primarily related to the fine-tuning approach, the amount and diversity of training data, or the underlying model capacity.